

Andrew Lee

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EDUCATION

Harvard University

Master of Science in Data Science

Cambridge, MA

Aug. 2025 – Dec. 2026 (Expected)

William and Mary

Bachelor of Science in Mathematics, Honors: Phi Beta Kappa

Williamsburg, VA

Aug. 2019 – May 2025

EXPERIENCE

Data Science Intern

E. & J. Gallo Winery

June 2025 – Aug. 2025

Modesto, CA

- Analyze data to transform into useful information to help make business decisions
- Data mining to identify patterns for predictive insights
- Organizing data using visualization
- Model structured and unstructured data for system integration

PROJECTS

Counting Cranes Using Computer Vision | *Computer Vision, MLOps* [Code]

Jan. 2024 – Dec. 2024

- Collaborated with U.S. Fish and Wildlife Service to develop deep learning models for counting migratory birds
- Trained YOLO-Family models on HPC infrastructure on 47,000+ imageries, achieving 5% average test error
- Built end-to-end MLOps workflow by packaging ML model in Docker and deploying to AWS Lambda via ECR and API Gateway
- Enabled weeks-long surveys to be completed in hours, accelerating policy decisions across conservation, hunting regulations, and funding allocations
- Presented research findings to the CEO of Conservation International and at multiple symposiums

Spatiotemporal Analysis of Wildfire Response | *R, Data Mining* [Code] [Paper]

June 2024 – Dec. 2024

- Conducted original research to model public awareness and behavior in response to wildfire smoke by forecasting interest in 10 health-related keywords using Google Trends
- Collected and processed 5 years of wildfire, weather, and socioeconomic data across 12 California DMAs using Selenium for automated web scraping
- Built a Bayesian spatiotemporal distributed lag model, revealing a 14% rise in air pollution searches and 19% rise in air purifier interest within two weeks of elevated smoke PM2.5

Machine Learning in Hypersonics | *Statistical Learning, Time-Series* [Code]

May 2023 – Aug. 2023

- Built supervised learning models – Linear Regression, Random Forest, SVM – to predict STLE location in hypersonic vehicles using sparse pressure sensor data
- Engineered time-series features to reduce sensor noise and improve predictions, increasing R-squared from 0.79 to 0.89 and reducing MSE by 77%
- Trained deep neural network on only 7 of 37 experimental runs, successfully generalizing to 900K+ test observations

World Happiness Score Visualization | *Tableau, Matplotlib, Plotly, HTML/CSS* [Code]

Jan. 2023 – May. 2023

- Designed and deployed an interactive website to visualize global trends in happiness scores using HTML, CSS, and JavaScript via GitHub Pages
- Built dynamic dashboards in Tableau and integrated them to highlight cross-country differences in happiness metrics across years
- Cleaned and wrangled data from the World Happiness Report using pandas to prepare it for both exploratory analysis and dashboard creation

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Matplotlib, seaborn, Plotly), SQL, R

Machine Learning: Supervised and Unsupervised Learning, Computer Vision, NLP, Statistical Modeling

Developer Tools: Git, Docker, AWS (EC2, S3, ECR, SageMaker, Lambda), DVC, Linux/Bash